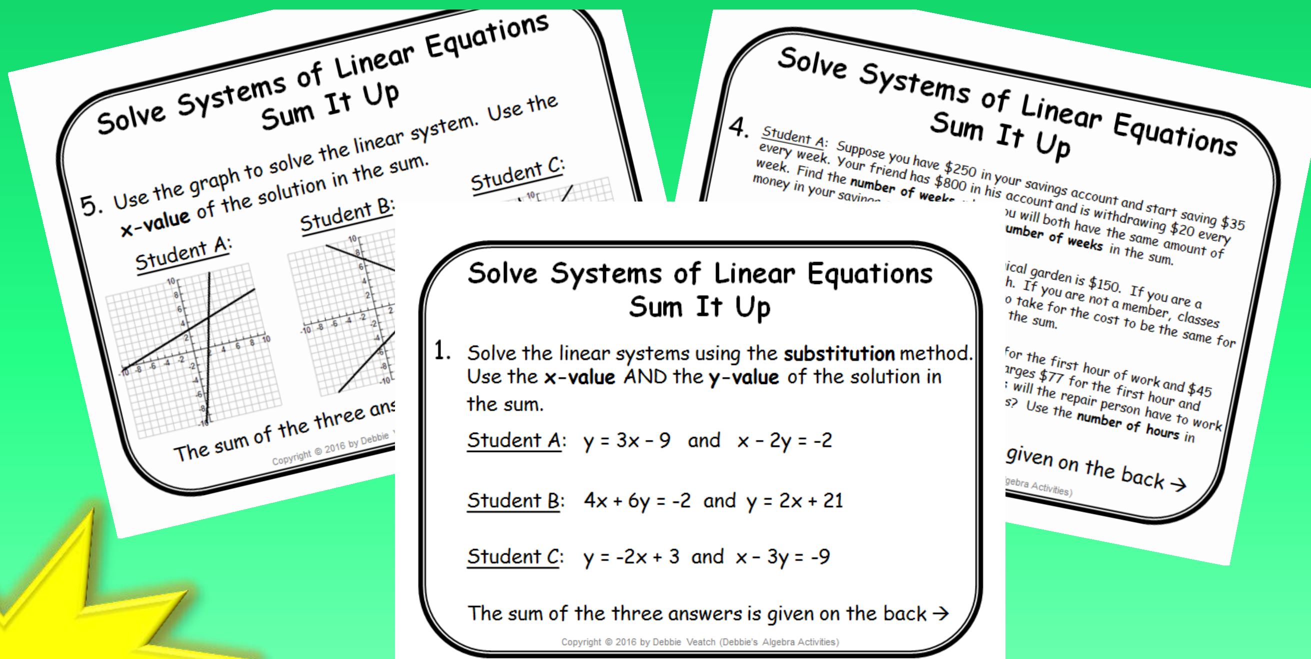
# SOLVE SYSTEMS OF LINEAR EQUATIONS


Solve Systems of Linear Equations
Sum It Up

5. Use the graph to solve the linear system. Use the x-value of the solution in the sum.

Solve Systems of Linear Equations
Sum It Up

Solve Systems of Linear Equations
Sum It Up

1. Solve the linear systems using the **substitution** method. Use the **x-value** AND the **y-value** of the solution in the sum.

Student A: $y = 3x - 9$ and $x - 2y = -2$

Student B: $4x + 6y = -2$ and $y = 2x + 21$

Student C: $y = -2x + 3$ and $x - 3y = -9$

The sum of the three answers is given on the back →

Copyright © 2016 by Debbie Veatch (Debbie's Algebra Activities)

Algebra 1

8 Stations

## Sum It Up Activity

Made by Debbie Veatch

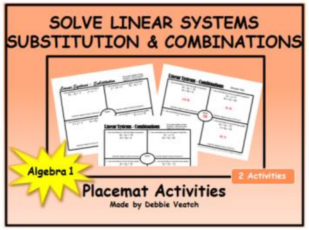
# Thank You!

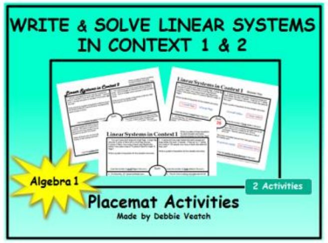
Thank you for downloading my product. I have personally used activities I have created in my own classroom and can testify to their effectiveness. I truly appreciate the support and look forward to your feedback.

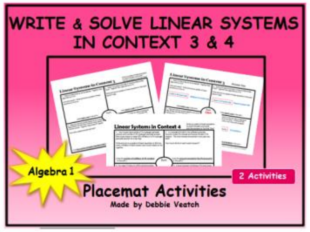
More activities are coming soon, so please check back often.

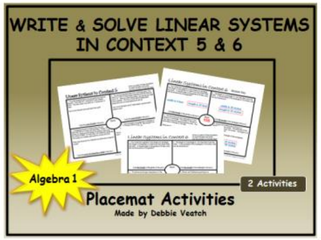
Debbie

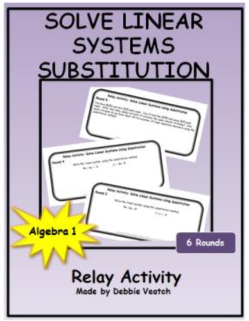
**You might also like:**

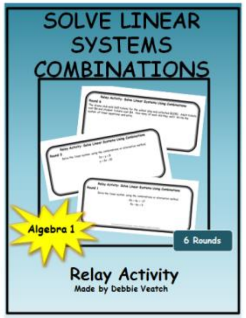



More activities coming soon . . .
Please check back often!

# Solve Systems of Linear Equations “Sum It Up” Activity

**Objective:**
To practice or review solving systems of linear equations using the graphing method, the substitution method and the linear combinations or elimination method.

**CCSS: Analyze and solve pairs of simultaneous linear equations.**

8.EE.C.8 Analyze and solve pairs of simultaneous linear equations.

8.EE.C.8a Understand that solutions to a system of two linear equations in two variables correspond to points of intersection of their graphs, because points of intersection satisfy both equations simultaneously.

8.EE.C.8b Solve systems of two linear equations in two variables algebraically, and estimate solutions by graphing the equations. Solve simple cases by inspection.

8.EE.C.8c Solve real-world and mathematical problems leading to two linear equations in two variables.

**CCSS: Solve systems of equations.**

HSA.REI.C5 Prove that, given a system of two equations in two variables, replacing one equation by the sum of that equation and a multiple of the other produces a system with the same solutions.

HSA.REI.C.6 Solve systems of linear equations exactly and approximately (e.g., with graphs), focusing on pairs of linear equations in two variables.

**CCSS: Create equations that describe numbers or relationships.**

HSA.CED.A.3 Represent constraints by equations or inequalities, and by systems of equations and/or inequalities, and interpret solutions as viable or nonviable options in a modeling context.

continued →

# Solve Systems of Linear Equations "Sum It Up" Activity

**Teacher Setup:**
Print each page on cardstock or paper. Write only the numerical "Sum" of the three answers on the back of the corresponding page for students to check. Attach the pages to the walls around the room as stations for the students to visit.

**Activity Directions:**
Students work in groups of three, labeling themselves Student A, Student B and Student C. Each student works their own problem at a station in their own notebook. Students then add their three answers together for the "Sum". They check the back of the station/page to see if the "Sum" of their answers is correct before moving to the next station. Students may rotate through the stations in any order.

**Included in the package:**
- Eight "Sum It Up" pages/stations, each with three problems, to put on walls around the classroom
- "Sum" answers to write on the back of each page/station for students to use
- Detailed answer key for the teacher

# Solve Systems of Linear Equations
# Sum It Up

1. Solve the linear systems using the **substitution** method. Use the **x-value** AND the **y-value** of the solution in the sum.

<u>Student A</u>: $y = 3x - 9$ and $x - 2y = -2$

<u>Student B</u>: $4x + 6y = -2$ and $y = 2x + 21$

<u>Student C</u>: $y = -2x + 3$ and $x - 3y = -9$

The sum of the three answers is given on the back →

# Solve Systems of Linear Equations
# Sum It Up

2. Solve the linear systems using the **linear combination** or **elimination** method. Use the **x-value** AND the **y-value** of the solution in the sum.

| Student A: | Student B: | Student C: |
|---|---|---|
| $2x - 3y = 8$ | $2x + 4y = 10$ | $5x + 2y = 9$ |
| $4x + 3y = 16$ | $-2x - 3y = -12$ | $-5x - 5y = 15$ |

The sum of the three answers is given on the back →

# Solve Systems of Linear Equations
# Sum It Up

3. Give the **number** you should **multiply** the first equation by to make the x's opposites for the linear combination or elimination method. Use this **number** in the sum.

Student A:

$x - 6y = -6$
$2x + 3y = 3$

Student B:

$x - 2y = 12$
$-3x + y = -1$

Student C:

$-x + 18y = 6$
$3x + 6y = 12$

The sum of the three answers is given on the back →

# Solve Systems of Linear Equations
# Sum It Up

4. <u>Student A</u>: Suppose you have $250 in your savings account and start saving $35 every week. Your friend has $800 in his account and is withdrawing $20 every week. Find the **number of weeks** when you will both have the same amount of money in your savings accounts. Use the **number of weeks** in the sum.

<u>Student B</u>: The cost to join the local botanical garden is $150. If you are a member, you can take classes for $7.50 each. If you are not a member, classes cost $20. **How many classes** will you have to take for the cost to be the same for both options? Use the **number of classes** in the sum.

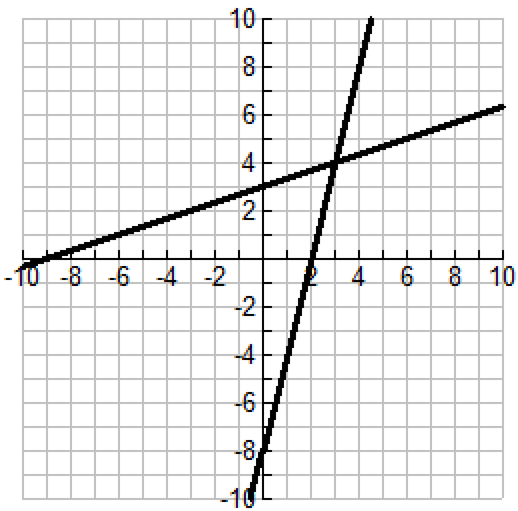
<u>Student C</u>: One repair company charges $35 for the first hour of work and $45 for each additional hour. Another company charges $77 for the first hour and $38 for each additional hour. **How many hours** will the repair person have to work for the cost to be the same with both companies? Use the **number of hours** in the sum.

The sum of the three answers is given on the back →

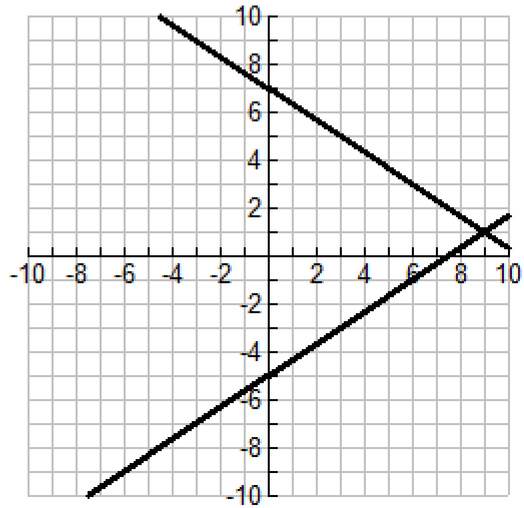
# Solve Systems of Linear Equations
# Sum It Up

5. Use the graph to solve the linear system. Use the **x-value** of the solution in the sum.

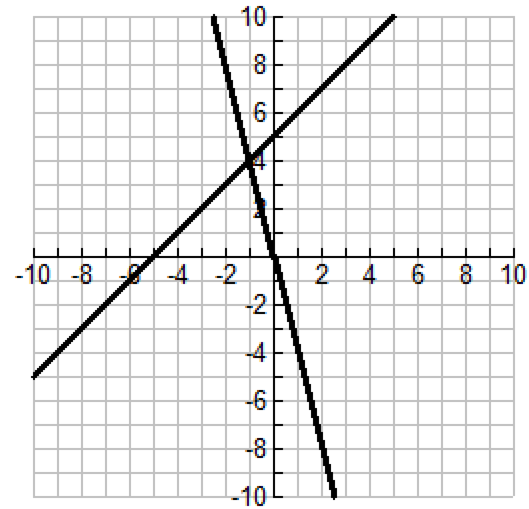
Student A:




Student B:




Student C:




The sum of the three answers is given on the back →

# Solve Systems of Linear Equations
# Sum It Up

6. Solve the linear systems using the **linear combination** or **elimination** method. Use the **x-value** AND the **y-value** of the solution in the sum.

<u>Student A</u>:

$4y = 3x - 21$

$4x + 2y = 6$

<u>Student B</u>:

$6x = -7y + 10$

$2x - 3y = 14$

<u>Student C</u>:

$3x = 10 + 2y$

$4x + 6y = -4$

The sum of the three answers is given on the back →

# Solve Systems of Linear Equations
# Sum It Up

7. Solve the linear systems using the **substitution** method. Use the **x-value** AND the **y-value** of the solution in the sum.

Student A: $2x - 3y = -5$ and $x + 2y = 1$

Student B: $x - 3y = -3$ and $2x + 6y = -6$

Student C: $x + 2y = 7$ and $3x - 2y = 5$

The sum of the three answers is given on the back →

# Solve Systems of Linear Equations
# Sum It Up

8. Solve the linear systems using the **linear combination** or **elimination** method. Use the **x-value** AND the **y-value** of the solution in the sum.

Student A:

$2x - 6y = -18$

$3x + 7y = 37$

Student B:

$3x + 2y = -5$

$4x - 3y = 16$

Student C:

$3x + 2y = 16$

$4x - 3y = -7$

The sum of the three answers is given on the back →

# Answers: Solve Systems of Linear Equations

Write these answers on the back of each corresponding page for students to check.

1. Sum is 7
2. Sum is 8
3. Sum is 4
4. Sum is 28
5. Sum is 11
6. Sum is 2
7. Sum is 2
8. Sum is 11

The next pages have detailed answers for the teacher only. *All answers should* **<u>not</u>** be written on the back of the stations/pages, only the "Sum" of the three answers.

Answer Key for Teacher

# Solve Systems of Linear Equations
# Sum It Up

1. Solve the linear systems using the **substitution** method. Use the **x-value** AND the **y-value** of the solution in the sum.

Student A: $y = 3x - 9$ and $x - 2y = -2$ $\boxed{(4, 3)}$

Student B: $4x + 6y = -2$ and $y = 2x + 21$ $\boxed{(-8, 5)}$

Student C: $y = -2x + 3$ and $x - 3y = -9$ $\boxed{(0, 3)}$

The sum of the three answers is given on the back →

**"Sum" = 7**

Answer Key for Teacher

# Solve Systems of Linear Equations
# Sum It Up

2. Solve the linear systems using the **linear combination** or **elimination** method. Use the **x-value** AND the **y-value** of the solution in the sum.

Student A:

$2x - 3y = 8$

$4x + 3y = 16$

$\boxed{(4, 0)}$

Student B:

$2x + 4y = 10$

$-2x - 3y = -12$

$\boxed{(9, -2)}$

Student C:

$5x + 2y = 9$

$-5x - 5y = 15$

$\boxed{(5, -8)}$

The sum of the three answers is given on the back →

**"Sum" = 8**

Answer Key for Teacher

# Solve Systems of Linear Equations
## Sum It Up

3. Give the **number** you should **multiply** the first equation by to make the x's opposites for the linear combination or elimination method. Use this **number** in the sum.

**Student A:**

$x - 6y = -6$ By $\boxed{-2}$

$2x + 3y = 3$

**Student B:**

$x - 2y = 12$ By $\boxed{3}$

$-3x + y = -1$

**Student C:**

$-x + 18y = 6$ By $\boxed{3}$

$3x + 6y = 12$

The sum of the three answers is given on the back →

$\boxed{\text{"Sum"} = 4}$

Answer Key for Teacher

# Solve Systems of Linear Equations
# Sum It Up

4. Student A: Suppose you have $250 in your savings account and start saving $35 every week. Your friend has $800 in his account and is withdrawing $20 every week. Find the **number of weeks** when you will both have the same amount of money in your savings accounts. Use the **number of weeks** in the sum. 10 weeks

Student B: The cost to join the local botanical garden is $150. If you are a member, you can take classes for $7.50 each. If you are not a member, classes cost $20. **How many classes** will you have to take for the cost to be the same for both options? Use the **number of classes** in the sum. 12 classes

Student C: One repair company charges $35 for the first hour of work and $45 for each additional hour. Another company charges $77 for the first hour and $38 for each additional hour. **How many hours** will the repair person have to work for the cost to be the same with both companies? Use the **number of hours** in the sum. 6 hours

The sum of the three answers is given on the back →

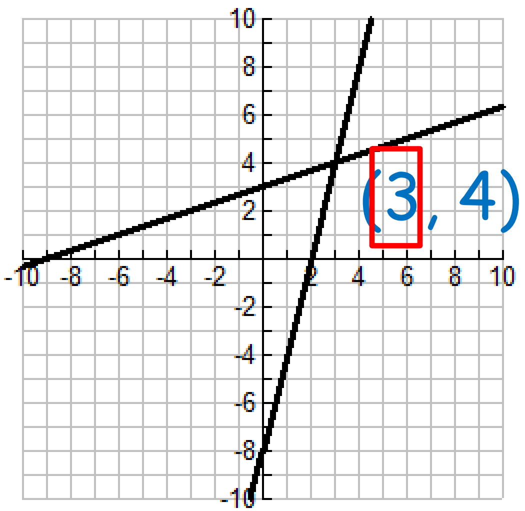
**"Sum" = 28**

Answer Key for Teacher

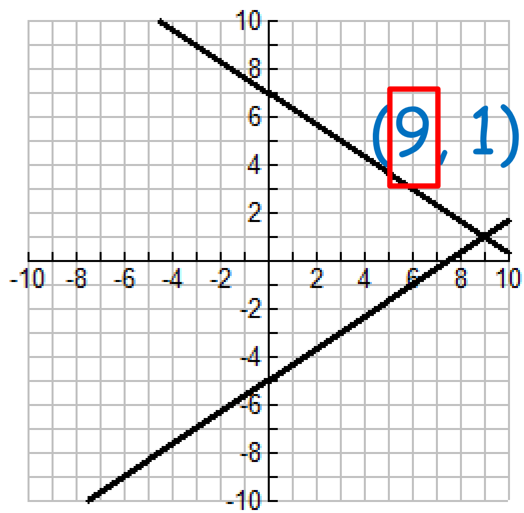
# Solve Systems of Linear Equations
# Sum It Up

5. Use the graph to solve the linear system. Use the **x-value** of the solution in the sum.

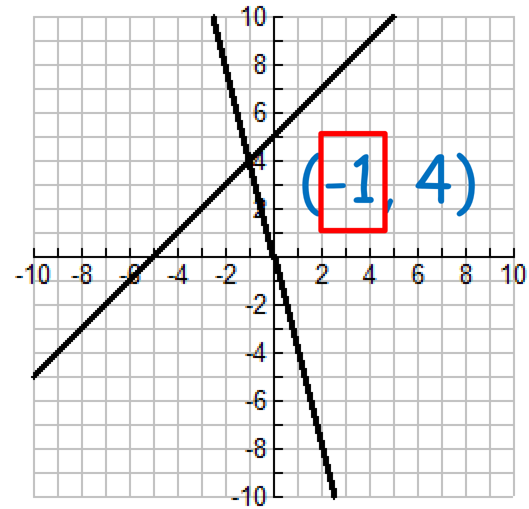
Student A:




Student B:




Student C:




The sum of the three answers is given on the back →

**"Sum" = 11**

Answer Key for Teacher

# Solve Systems of Linear Equations
# Sum It Up

6. Solve the linear systems using the **linear combination** or **elimination** method. Use the **x-value** AND the **y-value** of the solution in the sum.

Student A:

$4y = 3x - 21$

$4x + 2y = 6$

$\boxed{(3, -3)}$

Student B:

$6x = -7y + 10$

$2x - 3y = 14$

$\boxed{(4, -2)}$

Student C:

$3x = 10 + 2y$

$4x + 6y = -4$

$\boxed{(2, -2)}$

The sum of the three answers is given on the back →

**"Sum" = 2**

Answer Key for Teacher

# Solve Systems of Linear Equations
## Sum It Up

7. Solve the linear systems using the **substitution** method. Use the **x-value** AND the **y-value** of the solution in the sum.

Student A: $2x - 3y = -5$ and $x + 2y = 1$ $\boxed{(-1, 1)}$

Student B: $x - 3y = -3$ and $2x + 6y = -6$ $\boxed{(-3, 0)}$

Student C: $x + 2y = 7$ and $3x - 2y = 5$ $\boxed{(3, 2)}$

The sum of the three answers is given on the back →

$\boxed{\text{“Sum”} = 2}$

Answer Key for Teacher

# Solve Systems of Linear Equations
# Sum It Up

8. Solve the linear systems using the **linear combination** or **elimination** method. Use the **x-value** AND the **y-value** of the solution in the sum.

**Student A:**

$2x - 6y = -18$

$3x + 7y = 37$

$\boxed{(3, 4)}$

**Student B:**

$3x + 2y = -5$

$4x - 3y = 16$

$\boxed{(1, -4)}$

**Student C:**

$3x + 2y = 16$

$4x - 3y = -7$

$\boxed{(2, 5)}$

The sum of the three answers is given on the back →

**"Sum" = 11**